

Edition
5

Eckert

ANIMAL PHYSIOLOGY

*Mechanisms
and Adaptations*

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to the separation of an ultrafiltrate—a fluid devoid of colloidal protein particles—from blood plasma by filtration through a semipermeable membrane (the capillary wall) using pressure (blood pressure) to force the fluid through the membrane. Ultrafiltration occurs in most vertebrate kidneys, resulting in the net movement of a protein-free fluid from the blood into the kidney tubules. In general, capillary walls are permeable, so when blood pressures are high, fluid slowly filters across the walls and into the spaces between cells. In vertebrates, a lymphatic system has evolved to recover fluid lost to tissues from the blood in this manner and return it to the venous system. The extent of filtration depends largely on the blood pressure and the permeability of the capillary wall. The permeability of capillaries varies among tissues, and the blood pressure varies with circumstances and also with the organization of the vessels in the tissue. For example, in the liver, high permeability permits rapid transfer of substrates and products of metabolism, and pressures are lower than in the rest of the body. Low pressure in the lung capillaries reduces filtration into the gas space of the lungs, which would impair gas transfer.

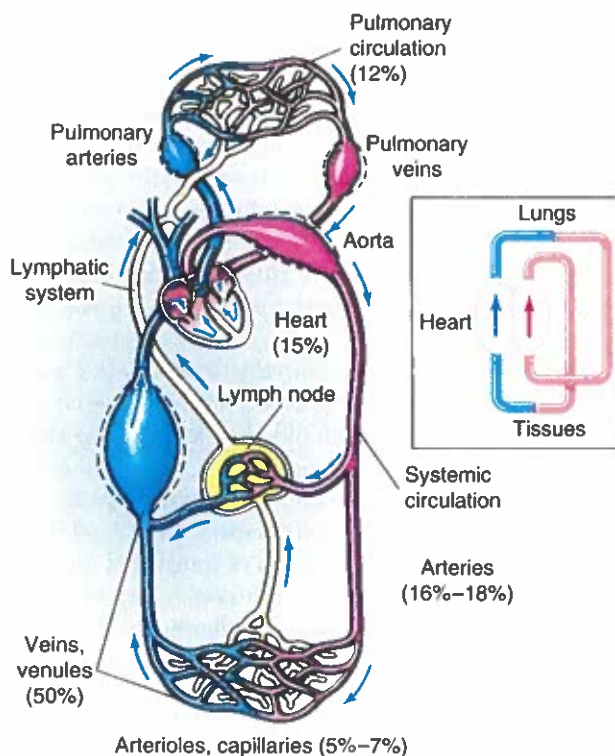


Figure 12-3 The closed circulation in mammals includes a fully divided heart, which permits different pressures to be maintained in the pulmonary and systemic circuits. The inset shows the general pattern of the mammalian circulation. The diagram illustrates the main components of the mammalian circulation, with oxygenated blood shown in red and deoxygenated blood shown in blue. The associated lymphatic system (yellow) returns fluid from the extracellular space to the bloodstream via the thoracic duct. The percentages indicate the relative proportion of blood in different parts of the circulation.

In vertebrates, the circulatory system is divided into a **systemic circuit** (which circulates blood throughout the body) and a **respiratory or pulmonary circuit** (which circulates blood to the organs of gas exchange). Mammals can maintain different pressures in these two circuits because they are equipped with a completely divided heart (Figure 12-3). The right side of the heart pumps blood through the pulmonary circuit, and the left side pumps blood through the systemic circuit. Note that the flows in the pulmonary and systemic circuits must be equal because blood returning from the lungs is pumped around the body. In other vertebrates, the heart is not completely divided, and flow to the lung can be varied independently of body blood flow.

The venous system collects blood from the capillaries and delivers it to the heart via the veins, which are typically low-pressure, flexible structures. Large changes in blood volume have little effect on venous pressure. Thus, the venous system contains most of the blood and acts as a large-volume reservoir. Blood donors give blood from this reservoir, and since there is little change in pressure as the venous volume decreases, the volumes and flows in other regions of the circulation are not markedly altered.

THE HEART

Start

Hearts are valved, muscular pumps that propel blood around the body. Hearts consist of one or more muscular chambers connected in series and guarded by valves or, in a few cases, sphincters (e.g., in some molluscan hearts), which allow blood to flow in only one direction. The mammalian heart has four chambers: two atria and two ventricles. Contractions of the heart result in the ejection of blood into the circulatory system. Multiple heart chambers permit stepwise increases in pressure as blood passes from the venous to the arterial side of the circulation (Figure 12-4).

Vertebrate cardiac muscle fibers are similar to skeletal muscles in many respects (see Chapter 10). Their distinguishing feature is the presence of gap junctions, which means that they are electrically coupled to one another. Except for differences in the uptake and release of Ca^{2+} , the mechanisms of contraction of vertebrate skeletal and cardiac muscle are generally similar. The myocardium (i.e., heart muscle) consists of three types of muscle fibers, which differ in size and functional properties:

- The myocardial cells in the sinus node (or sinoatrial node) and in the atrioventricular node are often smaller than others. They are autorhythmic, only weakly contractile, and exhibit very slow electrical conduction between cells.
- The largest myocardial cells, found in the inner surface of the ventricular wall, are also weakly contractile but are specialized for fast electrical

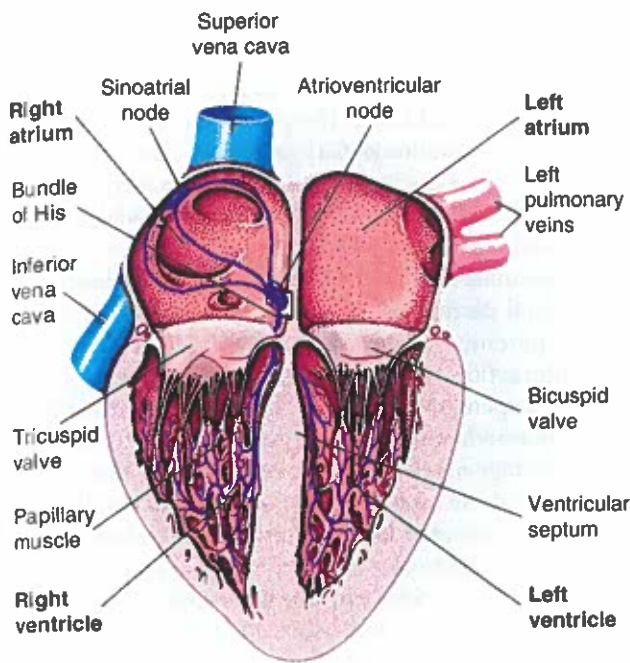


Figure 12-4 The multi-chambered mammalian heart permits the pressure to increase as blood moves from the venous to the arterial side. This cutaway view depicts the rear portion of the human heart, with the impulse pathways shown as blue lines. Impulses originate in the pacemaker region, located in the sinoatrial node, and spread to the atrioventricular node, from which they are transmitted to the ventricles. [Adapted from E. F. Adolph, 1967. Copyright © 1967 by Scientific American, Inc. All rights reserved.]

conduction and constitute the system for spreading excitation over the heart.

- The intermediate-sized myocardial cells are strongly contractile and constitute the bulk of the heart.



Why do you think so many species evolved a single large multi-chambered heart rather than a series of smaller hearts distributed throughout the circulation?

Electrical Properties of the Heart

A heartbeat consists of a rhythmic contraction (systole) and relaxation (diastole) of the whole cardiac muscle mass. The contraction of each cell is associated with an action potential (AP) in that cell. Electrical activity, initiated in the pacemaker region of the heart as described below, spreads over the heart from one cell to another because the cells are electrically coupled via gap junctions. The nature and extent of coupling determine the pattern by which the electrical wave of excitation spreads over the heart and also influence the rate of conduction.

Neurogenic and myogenic pacemakers

In fishes, the pacemaker is situated in the *sinus venosus* (see Figure 12-16); in other vertebrates, it is situated in a remnant of the sinus venosus called the sinoatrial node (see Figure 12-4). The pacemaker consists of a group of small, weakly contractile, specialized muscle cells that are capable of spontaneous activity. These cells may be either neurons, as in the neurogenic pacemaker in many invertebrate hearts, or muscle cells, as in the myogenic pacemaker in vertebrate and some invertebrate hearts. Hearts are often categorized by the type of pacemaker they have and hence are called either *neurogenic* or *myogenic* hearts.

In many invertebrates, it is not clear whether the pacemaker is neurogenic or myogenic. However, the hearts of decapod crustaceans (shrimps, lobsters, crabs) are plainly neurogenic. In these animals, the cardiac ganglion, situated on the heart, acts as a pacemaker. If the cardiac ganglion is removed, the heart stops beating, although the ganglion continues to be active and shows intrinsic rhythmicity. The cardiac ganglion consists of nine or more neurons (depending on the species), divided into small and large cells. The small cells act as pacemakers and are connected to large follower cells, which are all electrically coupled. Activity from the small pacemaker cells is fed into and integrated by the large follower cells and then distributed to the heart muscle. The crustacean cardiac ganglion is innervated by excitatory and inhibitory nerves originating in the central nervous system (CNS); these nerves can alter the rate of firing of the ganglion and therefore the rate at which the heart beats. Hormones released into the hemolymph can also modulate cardiac activity.

Vertebrate, molluscan, and many other invertebrate hearts are driven by myogenic pacemakers. These tissues have been studied extensively in a variety of species. A myogenic heart may contain many cells capable of pacemaker activity, but because all cardiac cells are electrically coupled, the cell (or group of cells) with the fastest intrinsic activity is the one that stimulates the whole heart to contract and determines the heart rate. These pacemaker cells normally overshadow those with slower pacemaker activity; however, if the normal pacemaker stops for some reason, the other pacemaker cells take over, producing a new, lower heart rate. Thus, cells with the capacity for spontaneous electrical activity may be categorized as pacemakers and latent pacemakers. In the event that a latent pacemaker becomes uncoupled electrically from the pacemaker, it may beat and control a portion of cardiac muscle—generally an entire chamber—at a rate different from that of the normal pacemaker. Such an *ectopic pacemaker* is dangerous because it desynchronizes the pumping action of the heart chambers.

Cardiac pacemaker potentials

An important characteristic of pacemaker cells is the absence of a stable resting potential. After each action

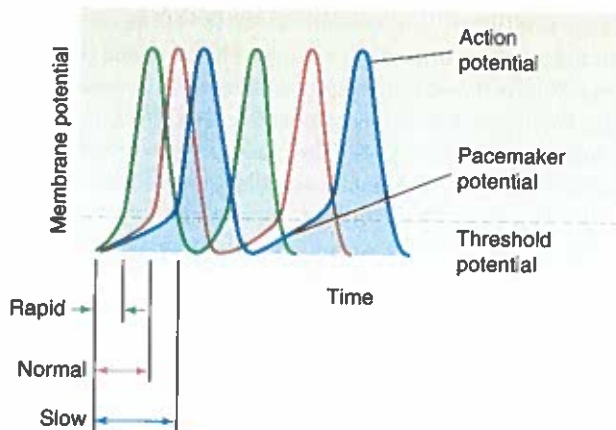


Figure 12-5 Pacemaker cells undergo a spontaneous depolarization of the membrane, referred to as the pacemaker potential, which triggers cardiac action potentials autorhythmically (red curve). A more rapid depolarization (green curve) increases the firing rate, and thus the heart rate, whereas a slower depolarization (blue curve) slows the firing rate and heart rate.

potential, the membrane of a pacemaker cell undergoes a steady depolarization, termed a *pacemaker potential*. The pacemaker potential brings the membrane to the threshold potential, usually in less than a second, giving rise to another all-or-none cardiac action potential. The interval between cardiac APs, which of course determines the heart rate, depends on the rate of the pacemaker potential, as well as on the extent of repolarization and the threshold potential for the cardiac AP. A faster pacemaker depolarization brings the membrane to a firing level sooner and thus increases the frequency of firing, leading to a faster heart rate, whereas a slower depolarization does the opposite (Figure 12-5).

In the frog sinus venosus, the pacemaker potential begins immediately after the previous AP, when the K^+ conductance of the membrane is high. The K^+ conduc-

tance then gradually drops, and the membrane shows a corresponding depolarization owing to the reduction in K^+ flux in the presence of a moderately high conductance for Na^+ and Ca^{2+} . The pacemaker potential continues until activation of Ca^{2+} and Na^+ channels causes the upstroke of the action potential. The cardiac AP is then produced by the completion of the Hodgkin cycle described in Chapter 5 (see Figure 5-24).

Pacemaker cells have high input impedance, so only small currents are needed to change their membrane potential. Pacemaker activity has its origin in the interaction between several time-dependent and voltage-dependent membrane currents in combination with time-independent background currents, with no single change in conductance accounting for the change in potential. At least six time- and voltage-dependent and ligand-operated K^+ channels, as well as several different Ca^{2+} and Na^+ channels, have been found in pacemaker cells. Some of the Ca^{2+} and K^+ channels also allow the passage of Na^+ . This complex array of channels allows the pacemaker cells to initiate the heartbeat and incorporates multiple safety factors. As a result, the pacemaker cells can function for many years without interruption.

Acetylcholine, which is released from parasympathetic terminals of the vagus nerve (Xth cranial nerve) that innervate the heart, slows the heartbeat by increasing the K^+ conductance and reducing the Ca^{2+} conductance of pacemaker cells. The increase in K^+ conductance keeps the membrane potential near the K^+ equilibrium potential for a longer time, thereby slowing the pacemaker potential and delaying the onset of the next upstroke (Figure 12-6a). Adenosine also slows the heart via a mechanism similar to that of acetylcholine by modifying K^+ conductance. In contrast, norepinephrine released from sympathetic nerves accelerates the pacemaker potential, thus increasing the heart rate (Figure 12-6b). Catecholamines, such as

Neurotransmitters, acetylcholine (ACh), etc.- remember for toad heart lab

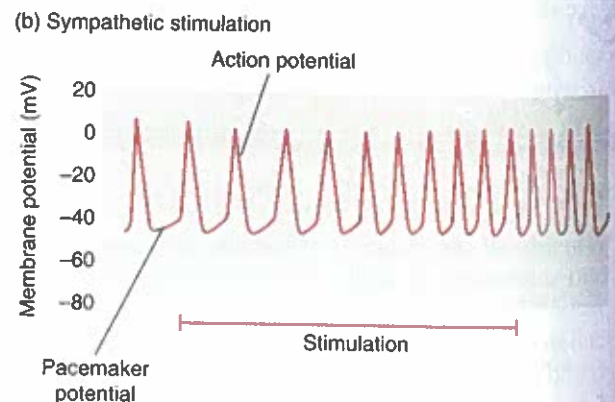
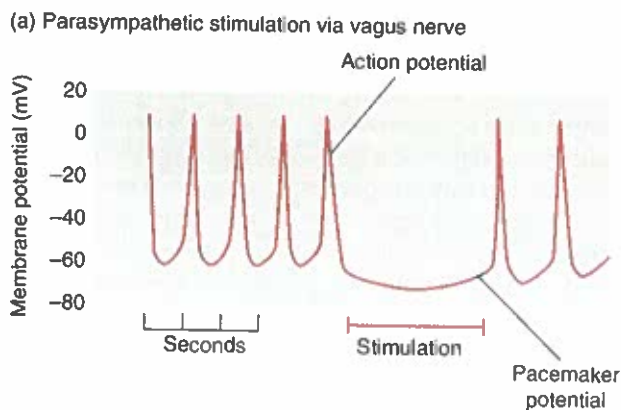


Figure 12-6 Parasympathetic stimulation via the vagus nerve and sympathetic stimulation have opposite effects on the pacemaker potential and heart rate. (a) Vagus stimulation produces a rise in diastolic (resting) transmembrane potential, a decrease in the rate of

depolarization, and a decrease in the duration and frequency of the action potential. (b) Sympathetic stimulation produces an increase in the frequency of firing of the pacemaker cells. [From Hutter and Trautwein, 1956.]

norepinephrine, bind to β -adrenergic receptors on the cell surface, activating adenylate cyclase and increasing the levels of cyclic AMP (cAMP) in the pacemaker cell. The increase in cAMP activates cation channels triggered by Ca^{2+} , K^{+} , and hyperpolarization, as well as the $\text{Na}^{+}/\text{K}^{+}$ pump. The overall effect is a more rapid depolarization of the pacemaker cell and an increase in heart rate. Some of these same channels can be activated by an increase in the level of phosphorylation, which increases the intrinsic heart rate and results in a greater sensitivity to β -adrenergic agonists.

Cardiac action potentials

The action potentials that precede contraction in all vertebrate cardiac muscle fibers are of longer duration than those in skeletal muscle. In skeletal muscle, the AP is completed and the membrane is in a nonrefractory state before the onset of contraction; hence, repetitive stimulation and tetanic contraction are possible (Figure 12-7a). In cardiac muscle fibers, however, the AP reaches a plateau at which it remains for hundreds of milliseconds (Figure 12-7b), and the membrane remains in a refractory state until the heart has returned to a relaxed state. Thus, summation of contractions does not occur in cardiac muscle.

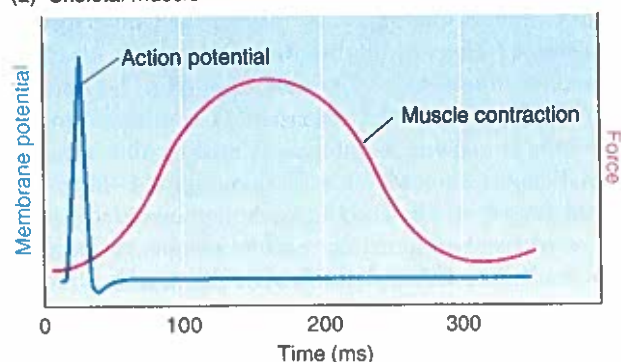
In most cardiac fibers, APs begin with a rapid depolarization that results from a large and rapid increase in Na^{+} conductance. This depolarization differs from the slow depolarization of the pacemaker potential, which depends on a stable Na^{+} conductance and decreasing K^{+} conductance. Repolarization of the plasma membrane is delayed for hundreds of milliseconds during the so-called plateau phase (see Figure 12-7b). The long duration of the cardiac AP produces a prolonged contraction, so that an entire heart chamber can fully contract before any portion begins to

relax—a process that is essential for efficient pumping of blood.

The prolonged plateau of the cardiac AP results from the maintenance of a high Ca^{2+} conductance and a delay in the subsequent increase in K^{+} conductance (unlike the situation in skeletal muscle). The inflow of Ca^{2+} is especially important in lower vertebrates, in which a considerable proportion of the Ca^{2+} essential for activation of contraction enters through the plasma membrane. In birds and mammals, the surface-to-volume ratio of the larger cardiac muscle cells is too small to allow sufficient entry of Ca^{2+} to fully activate contraction. Most of the Ca^{2+} required to trigger contraction is released from the sarcoplasmic reticulum (extensive in cardiac muscle fibers of higher vertebrates) by calcium-mediated calcium release (see Chapter 10). A rapid repolarization, due to a fall in Ca^{2+} conductance and an increase in K^{+} conductance, terminates the plateau phase.

The duration of the plateau phase and the rates of depolarization and repolarization vary among different cells of the same heart. The summation of these changes can be recorded by an electrocardiogram (ECG). Because of the large number of cells involved, the currents that flow during the synchronous activity of cardiac cells can be detected as small changes in potential from points all over the body, and can be easily recorded and then analyzed. Such a recording shows a characteristic pattern of electrical activity (Figure 12-8a on the next page). The initial P-wave is associated with depolarization of the atrium, the so-called QRS complex with depolarization of the ventricle, and the T-wave with repolarization of the ventricle. The electrical activity associated with atrial repolarization is obscured by the much larger QRS complex. The exact form of the electrocardiogram varies by species and is affected by

(a) Skeletal muscle



(b) Cardiac muscle

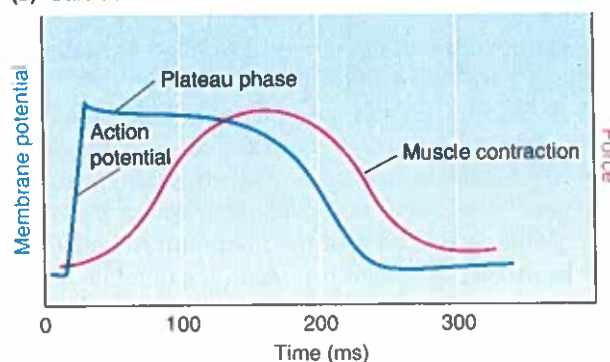


Figure 12-7 Action potentials differ in skeletal and cardiac muscle. (a) Action potentials in skeletal muscle are of very short duration. (b) In contrast, cardiac action potentials exhibit a prolonged repolarization, or plateau phase, during

which the muscle fiber is refractory to stimulation. For this reason, repetitive stimulation during a contraction and summation of contractions can occur in the skeletal muscle, but not in cardiac muscle.

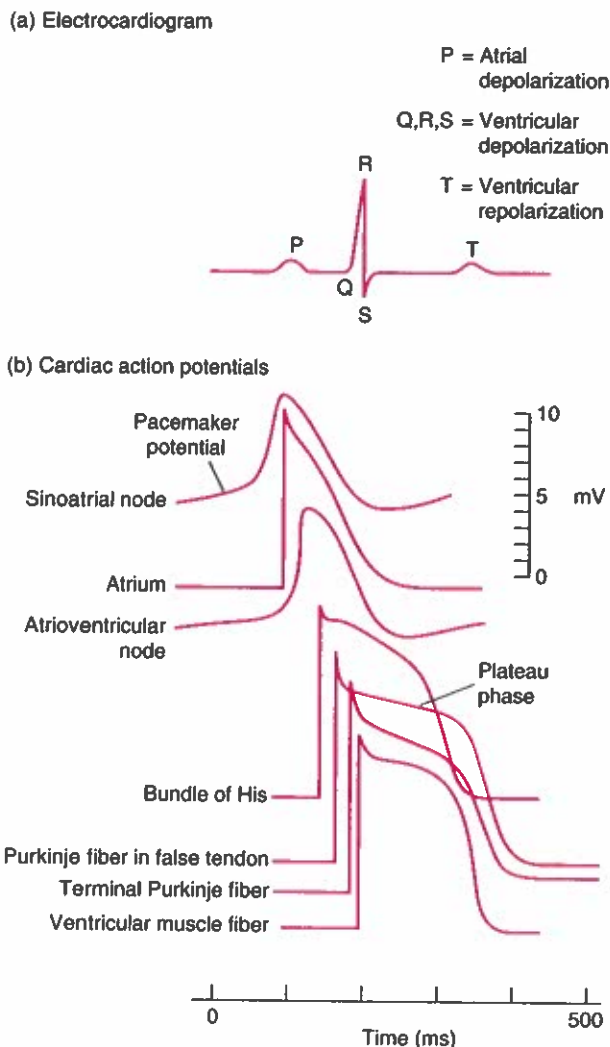


Figure 12-8 An electrocardiogram represents the summation of the electrical activity in various parts of the heart. (a) The major components of the electrocardiogram reflect atrial depolarization (P), ventricular depolarization (QRS), and ventricular repolarization (T). (b) The amplitude, configuration, and duration of cardiac action potentials differ at various sites in the heart. The sites shown fire in sequence from top to bottom. [Part b from Hoffman and Cranefield, 1960.]

the nature and position of the recording electrodes, as well as by the nature of cardiac contraction.

Atrial cells generally have an AP of shorter duration than ventricular cells (Figure 12-8b). The duration of the AP in atrial or ventricular cells from the hearts of different species also varies. The duration of the AP is one factor correlated with the maximum frequency of the heartbeat; in smaller mammals, the duration of the ventricular AP is shorter, and heart rates are higher, than in larger mammals.

Because of the great diversity among the hearts of different invertebrates, few generalizations can be made about the ionic mechanisms that generate the cardiac APs of these phyla. The one widespread characteristic is participation of Ca^{2+} in the initiation of the action potential.

Transmission of excitation over the heart

Electrical activity initiated in the pacemaker region is conducted over the entire heart as depolarization in one cell results in the depolarization of neighboring cells by virtue of current flow through gap junctions. These junctions are situated where neighboring myocardial cells are closely apposed, in a region termed the intercalated disk (see figure 10-48). The adhesion of cells at intercalated disks is strengthened by the presence of anchoring structures called desmosomes. The area of contact is increased by folding and interdigitation of the plasma membranes, the extent of which increases during development of the heart. Gap junctions allow current to flow with minimal resistance from one cell to the next across intercalated disks.

Although the junctions between myocardial cells can conduct electric current in both directions, transmission is generally unidirectional because the impulse spreads away from the pacemaker region. There are usually several pathways for the excitation of any single cardiac muscle fiber, since intercellular connections are numerous. If a portion of the heart becomes nonfunctional, the wave of excitation can easily flow around that portion, so that the remainder of the heart can still be excited. The prolonged nature of cardiac APs ensures that these multiple connections do not result in multiple stimulation and a reverberation of activity in cardiac muscle. An AP initiated in the pacemaker region results in a single AP being conducted through all the other myocardial cells, and another AP from the pacemaker region is required for the next wave of excitation.

In the mammalian heart, the wave of excitation begins in the sinoatrial node and spreads over both atria in a concentric fashion at a velocity of about $0.8 \text{ m} \cdot \text{s}^{-1}$. The atria are connected electrically to the ventricles only through the atrioventricular node on the right side of the heart; in other regions, the atria and ventricles are joined by connective tissue that does not conduct the wave of excitation (see Figure 12-4). Excitation spreads to the ventricle through small junctional fibers, in which the velocity of the wave of excitation is slowed to about $0.05 \text{ m} \cdot \text{s}^{-1}$. The junctional fibers are connected to nodal fibers, which in turn are connected via transitional fibers to the bundle of His. This structure branches into right and left bundles, which subdivide into Purkinje fibers that extend into the myocardium of the two ventricles. Conduction is slow through the nodal fibers (about $0.1 \text{ m} \cdot \text{s}^{-1}$), then rapid ($4\text{--}5 \text{ m} \cdot \text{s}^{-1}$) as the bundle of His and the Purkinje fibers deliver the wave of excitation to all regions of the ventricular myocardium, causing all the ventricular muscle fibers to contract together. As each wave of excitation arrives, the ventricular myocardial cells contract almost immediately, with the wave of excitation passing at a velocity of $0.5 \text{ m} \cdot \text{s}^{-1}$ from the internal lining of the heart wall (endocardium) to the external covering (epicardium). The functional significance of the electrical organization of the myocardium is its ability to generate sepa-

rate, synchronous contractions of first the atria and then the ventricles. Thus, slow conduction through the atrioventricular node allows atrial contractions to precede ventricular contractions, providing time for blood to move from the atria into the ventricles.

As noted earlier, acetylcholine (ACh), released from parasympathetic nerve fibers, increases the interval between APs in pacemaker cells and thus slows the heart rate (see Figure 12-6a). This decrease in heart rate is sometimes referred to as a *negative chronotropic effect*. Parasympathetic cholinergic fibers in the vagus nerve innervate the sinoatrial node and atrioventricular node of the vertebrate heart. As the heart rate slows, acetylcholine also reduces the velocity of conduction from the atria to the ventricles through the atrioventricular node. High levels of acetylcholine block transmission through the atrioventricular node, so that only every second or third wave of excitation is transmitted to the ventricle. Under these unusual conditions, the atrial contraction rate will be two or three times that in the ventricle. Alternatively, high levels of acetylcholine may completely block conduction through the atrioventricular node (called *atrioventricular block*), allowing an ectopic pacemaker in the ventricle to take over. In this situation, the atria and ventricles are controlled by different pacemakers and contract at quite different rates, with the two beats uncoordinated. This event would be devastating for a fish, in which atrial contraction is very important for ventricular filling. It is not quite so devastating in mammals because atrial contraction only tops up the ventricles; they are filled mainly by the direct inflow of blood from the venous system through the relaxed atria.

The catecholamines epinephrine and norepinephrine have three distinct positive effects on heart function:

- They increase the rate of myocardial contractions, or heart rate (*positive chronotropic effect*).
- They increase the force of myocardial contraction (*positive inotropic effect*).
- They increase the speed of conduction of the wave of excitation over the heart (*positive dromotropic effect*).

The effect of epinephrine and norepinephrine on the rate of contraction is mediated via the pacemaker, whereas the increased strength of contraction is a general effect on all myocardial cells. Norepinephrine also increases conduction velocity through the atrioventricular node. It is released from adrenergic nerve fibers that innervate the sinoatrial node, atria, atrioventricular node, and ventricles, so that sympathetic adrenergic stimulation has a direct effect on all portions of the heart.

Mechanical Properties of the Heart

The vertebrate heart is a highly sophisticated pump that operates throughout the life of the animal, which in some instances exceeds 100 years. It must meet the

requirements of the body for blood flow whether the animal is at rest, upside down, or exerting itself violently.

The chambers of the heart are not in a straight line, but are usually arranged in a looped pattern (Figure 12-9). The vessels delivering blood to and taking blood away from the heart are also usually curved, rather than

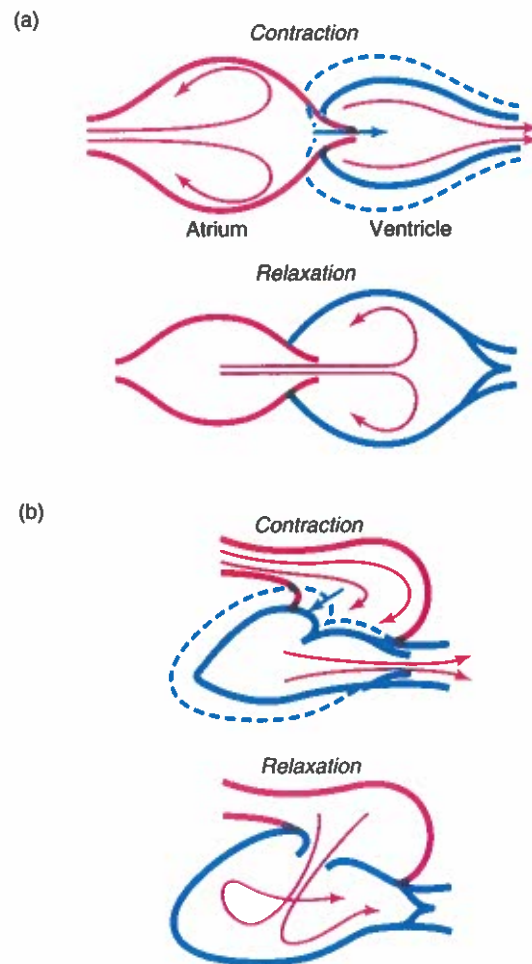


Figure 12-9 A looped atrioventricular arrangement enhances blood flow through the heart. (a) In heart chambers arranged linearly, contraction (systole) of the ventricle would pull on the atrium (blue arrow), contributing to atrial expansion. The flow of blood into the atrium would give rise to bilateral recirculation (curved arrows), a relatively unstable pattern of flow that would redirect blood inappropriately for subsequent ventricular filling. Any recoil of the ventricle away from the blood being expelled from it would push back against the atrium, counteracting atrial expansion. During relaxation (diastole), recirculation in the expanding ventricle would redirect fluid away from the outflow tract. (b) In a sinusoidally looped arrangement of heart chambers, ventricular contraction expands the adjacent atrium (blue arrow). Atrial filling is stable and streamlined, accommodated by wall curvatures that redirect momentum toward the atrioventricular valve. Recoil of the ventricle away from ejected blood now adds to the pull on the atrioventricular junction, enhancing rather than suppressing atrial expansion. Redirected blood flow in the atrium contributes to ventricular filling and, in turn, ventricular inflow is redirected toward the outflow tract. [Adapted from P. H. Kilner, G.-Z. Yang, J. A. Wilkes, R. H. Mohiaddin, D. N. Firmin, and M. H. Yacoub, 2000.]

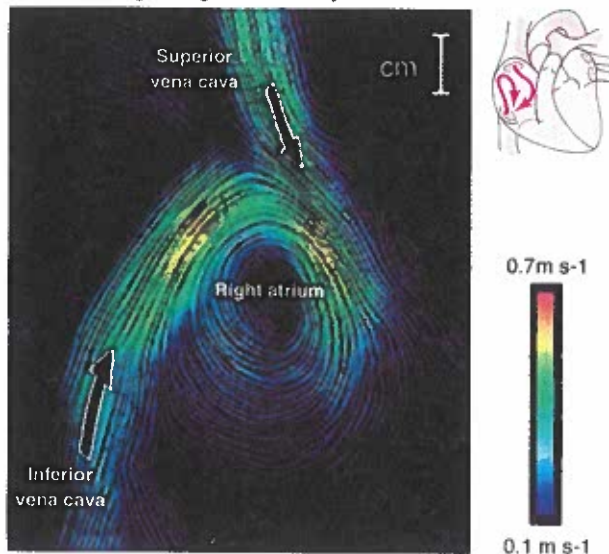
straight. In addition, many hearts have grooves and ridges within the chambers and vessels. These features allow the contraction of one chamber to facilitate blood movement in other chambers. They also create complex flow patterns that direct and maintain momentum, reducing the energy spent by the heart in generating flow. They are found in all classes of vertebrates, including several amphibians and many reptiles. Figure 12-10 shows the patterns of flow found in the human heart. These patterns minimize collisions between the streams of blood entering the heart and maintain momentum in the blood, directing it toward the next destination. The

blood vessels are curved to receive the swirling blood ejected from the ventricle without disrupting flow.

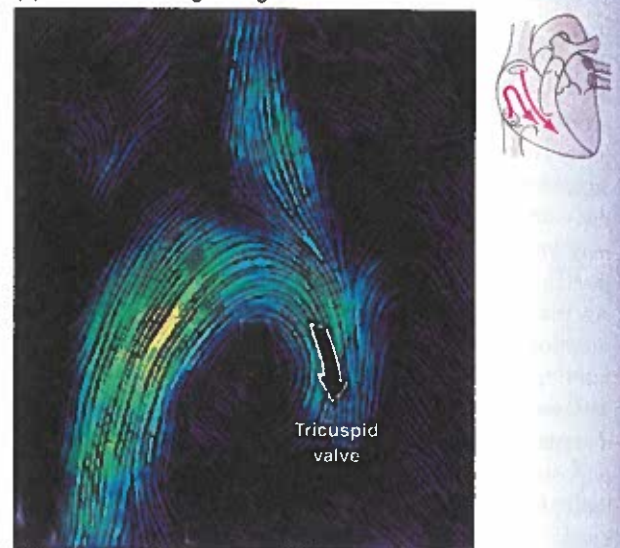
Heart rate, stroke volume, and cardiac output

Cardiac output is the volume of blood pumped per unit time from a ventricle. In mammals, it is defined as the volume ejected from the right or left ventricle, not the combined volume from both ventricles. The volume of blood ejected from a ventricle by each beat of the heart is termed the **stroke volume**. The mean stroke volume can be determined by dividing cardiac output by heart rate.

(a) Atrial filling during ventricular systole



(b) Ventricular filling during diastole



(c) Ventricular systole



(d) Ventricular diastole



Figure 12-10 Blood flow patterns through the human heart. (a) In ventricular systole, blood entering from superior and inferior venae cavae contributes to the forward rotation of blood in the expanding right atrium. Colored streamlines computed by magnetic resonance imaging show local speed, as indicated by the color scale. (b) In early ventricular diastole, further inflow of blood is again redirected forward and down the front of the right atrium, and away from the viewer (out of plane) through the open tricuspid valve. (c) In the left ventricle

in systole, streamlines pass from the left ventricle through the aortic valve, viewed here in an oblique long-axis plane from above left, the front of the subject being located to the left of the image. (d) In early diastole, streamlines pass from the left atrium through the open mitral valve to the left ventricle, with asymmetric recirculation (curved arrow) round the anterior leaflet of the mitral valve. In c and d only, the color scale is modified to reach red at $1 \text{ m} \cdot \text{s}^{-1}$. [From Kilner, Yang, Wilkes, Mohiaddin, Firmin, and Yacoub, 2000.]

Stroke volume is the difference between the volume of blood in the ventricle just before contraction (end-diastolic volume) and the volume in the ventricle at the end of a contraction (end-systolic volume). Changes in stroke volume may therefore result from changes in either end-diastolic or end-systolic volume. The end-diastolic volume is determined by four parameters:

- venous filling pressure
- pressures generated during atrial contraction
- distensibility of the ventricular wall
- the time available for filling the ventricle

The end-systolic volume is determined by these two parameters:

- pressures generated during ventricular contraction

- the pressure in the outflow channels from the heart (the aortic and pulmonary arteries)

In an isolated mammalian heart, increasing the venous filling pressure causes an increase in end-diastolic volume and results in an increased stroke volume (Spotlight 12-2). End-systolic volume also increases, but not as much as end-diastolic volume. Thus, cardiac muscle behaves in a way similar to skeletal muscle in that stretch of the relaxed muscle within a certain range of length results in the development of increased tension during a contraction. Increases in arterial pressure also cause a rise in both end-diastolic and end-systolic volume, but with little change in stroke volume. In this instance, the increased mechanical work required to maintain stroke volume in the face of elevated arterial pressure results from the increased stretch of cardiac muscle during diastole.

Spotlight 12-2

THE FRANK-STARLING MECHANISM

Remember for Toad Heart lab
(and muscle labs!)

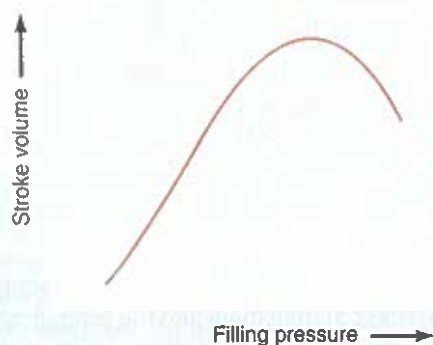
Otto Frank observed that the more a frog heart was filled by venous return, the greater the stroke volume. Frank derived a length-tension relationship for the frog myocardium and demonstrated that its contractile tension increases with stretch up to a maximum, then decreases with further stretch. Ernest Starling, a dominant figure in many areas of physiology during the early 1900s, had come to similar conclusions. Although neither Starling nor Frank considered mechanical work in their calculations, the increase in mechanical work by the ventricle caused by an increase in end-diastolic volume (in this case, due to venous filling pressure) is termed the Frank-Starling mechanism (see part a of the accompanying figure). The curves derived from measuring work output from the ventricle at different venous filling pressures are known as Starling curves (part b of the figure).

No single Starling curve, however, describes the relationship between venous filling pressure and work output from the

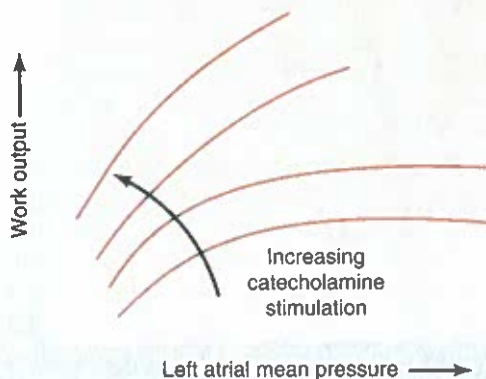
ventricle. The mechanical properties of the heart are affected by a number of factors, including the level of activity in the nerves innervating the heart and the composition of the blood perfusing the myocardium. For instance, the relationship between ventricular work output and venous filling pressure is markedly affected by stimulation of sympathetic nerves innervating the heart.

Starling was a versatile researcher who, along with William Bayliss, discovered the hormone secretin. He coined the term *hormones* and defined their basic properties (see Chapter 9). Starling also made many contributions to our understanding of the circulation. In addition to the observations described by the Frank-Starling mechanism, he proposed the Starling hypothesis, which states that the exchange of fluid between blood and tissues is due to differences in blood pressure and colloid osmotic pressures across the capillary wall (see Figure 12-38). This hypothesis was subsequently confirmed, largely by the work of E. Landis.

(a) Frank-Starling mechanism in frog heart



(b) Starling curves (measured in a mammalian heart)



As noted above, epinephrine and norepinephrine increase the force of contraction of the ventricles; hence both the rate and the extent of ventricular emptying are increased by these catecholamines. The adrenergic sympathetic nerves have a markedly greater effect on the rate and force of ventricular output during each heartbeat than does the cholinergic contribution of the vagus nerve. This difference stems from the much more extensive innervation of the ventricles by adrenergic nerves than by cholinergic nerves.

The effects of sympathetic nerve stimulation or increased circulating levels of catecholamines represent a series of integrated actions. Stimulation of pacemaker cells leads to an increase in heart rate. The velocity at which the wave of excitation is conducted across the heart is increased throughout the heart to produce a more nearly synchronous beat of all parts of the ventricle. The production and consumption of ATP in ventricular cells increases, leading to an increase in ventricular work: the rate of ventricular emptying during systole increases, so that the same or a larger stroke volume is ejected in a much shorter time. This increased force of contraction is mediated by the action of catecholamines on both α - and β -adrenoreceptors (see Chapter 9 for more details). Thus, when adrenergic nerve stimulation increases the heart rate, the same stroke volume is ejected from the heart in a shorter time. Even though the time available for emptying and filling the heart is reduced as heart rate increases, stroke volume remains quite stable over a wide range of heart rates. For example, exercise in many mammals is associated with a large increase in heart rate with little change in stroke volume; only at the highest heart rates does the stroke volume fall (Figure 12-11). Stroke volume is relatively invariant because, over a wide range of heart rates,

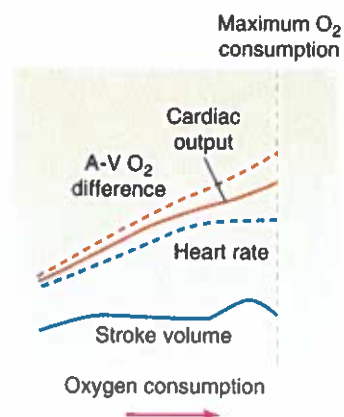


Figure 12-11 In humans and many other mammals, the increased requirements for cellular oxygen during exercise are met in part by increasing heart rate rather than stroke volume, leading to higher cardiac output. At high levels of oxygen consumption, heart rate levels off and stroke volume increases; at still higher levels, stroke volume decreases. In addition, extraction of oxygen from the blood in the capillaries increases during exercise, as indicated by the increase in the arterial-venous (A-V) O_2 difference.

increased sympathetic activity ensures more rapid ventricular emptying, and elevated venous pressures result in more rapid ventricular filling as heart rate increases.

There are limits to the amount by which diastole can be shortened, determined by the maximum possible rate at which the ventricles can be filled and emptied as well as by the nature of the coronary circulation (the circulation within the heart itself). Although it may seem paradoxical, coronary capillaries are occluded during contractions of the myocardium, so coronary blood flow is greatly reduced during systole. Coronary flow rises dramatically during diastole, but a decrease in the diastolic period reduces the period of flow.

Following experimental sympathetic denervation of the heart, exercise still results in increased cardiac output, but in this case there are large changes in stroke volume rather than in heart rate. The increased cardiac output is probably caused by an increase in venous return. It thus appears that sympathetic nerves are not involved in increasing cardiac output *per se*, but rather in raising heart rate and maintaining stroke volume, avoiding the large pressure oscillations associated with large stroke volumes and keeping the heart operating at or near its optimal stroke volume for efficiency of contraction. The sympathetic nerves thus play an important role in determining the relation between heart rate and stroke volume, but additional factors are involved in mediating the increase in cardiac output with exercise.

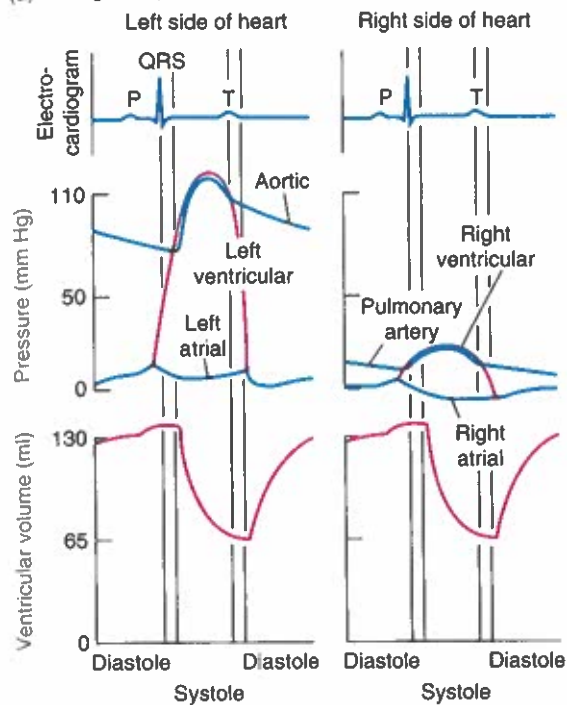
Cardiac regulation is typical of invertebrates too. For example, heart rate and stroke volume can be controlled independently in crabs and lobsters. In crustaceans, hemolymph leaves the heart via several arteries, and blood distribution is tightly regulated by cardio-arterial valves at the base of each artery. These muscular valves are innervated and are under both neuronal and neurohormonal control. Cardiac output in lobsters and crayfishes has been shown to increase with exercise, and adenosine, which stimulates heart rate and hemolymph flow in crayfishes, may be involved in the cardiovascular changes associated with exercise.

Changes in pressure and flow during a single heartbeat

Contractions of the mammalian heart cause fluctuations in cardiac pressure and volume, as illustrated by the tracings in Figure 12-12a. The following sequence of events occurs during a cardiac cycle (Figure 12-12b):

1. During diastole, closed aortic valves maintain large pressure differences between the relaxed ventricles and their outflow channels, the systemic aorta and pulmonary artery. The atrioventricular valves are open, and blood flows directly from the venous system into the ventricles.
2. When the atria contract, the pressures within them rise, and blood is ejected from them into the ventricles.
3. As the ventricles begin to contract, ventricular pressures rise and exceed those in the atria. At this point,

(a) Changes in pressure and volume during heartbeat



(b) Sequence of events in heartbeat

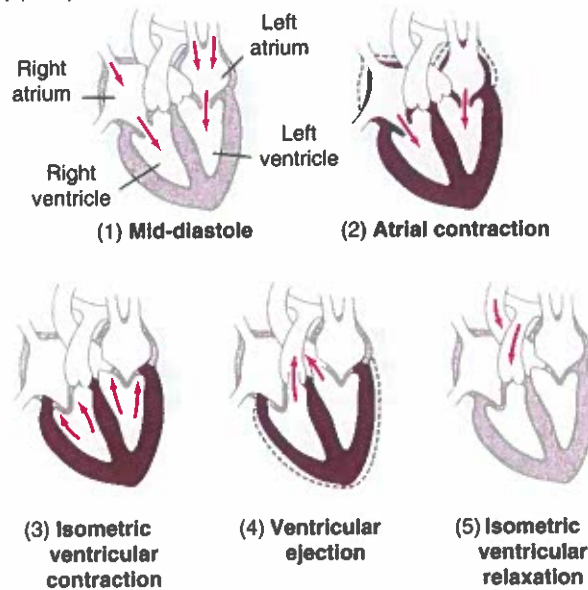


Figure 12-12 During a single cardiac cycle, sequential contraction of the atria and ventricles and the opening and closing of valves produce characteristic changes in pressure and volume. (a) Changes in pressure and volume in the

ventricles and aorta (left) and pulmonary artery (right) during a single cardiac cycle. (b) Sequence of events in contraction of the mammalian heart. Dark red indicates contracted muscle.

the atrioventricular valves close, preventing backflow of blood into the atria, and ventricular contraction proceeds. During this phase, both the atrioventricular and the aortic valves are closed, so that the ventricles form sealed chambers, and there is no volume change. That is, the ventricular contraction is isometric.

4. Pressures within the ventricles increase rapidly and eventually exceed those in the systemic and pulmonary aortas. The aortic valves are then pushed open, and blood is ejected into the aortas, resulting in a decrease in ventricular volume.
5. As the ventricles begin to relax, intraventricular pressures fall below the pressures in the aortas, the aortic valves close, and there is an isometric relaxation of the ventricle.

Once the ventricular pressures fall below those in the atria, the atrioventricular valves are pushed open, ventricular filling starts again, and the cycle is repeated.

In the mammalian heart, the volume of blood forced into the ventricle by atrial contraction is about 30% of the volume of blood ejected into the aorta by ventricular contraction. Thus, ventricular filling is largely determined by the venous filling pressure, which forces blood from the venous system directly through the atria into the ventricles. Atrial contraction simply tops off the nearly full ventricles. Still, maximal cardiac

output may be compromised if atrial contraction is impaired.

The contraction of cardiac muscle can be divided into two phases. The first is an isometric contraction, during which tension in the muscle and pressure in the ventricle increase rapidly. The second phase is essentially isotonic; tension does not change very much, for as soon as the aortic valves open, blood is ejected rapidly from the ventricles into the arterial system with little increase in ventricular pressure. Thus, tension is generated first with almost no change in muscle length; then the muscle shortens with little change in tension.

Work done by the heart

It is a simple principle of physics that mechanical work done is the product of mass times distance moved. In the context of the heart, work can be calculated as the change in pressure times flow. Flow is directly related to the change in volume with each contraction of the ventricle. With pressure given in grams per square centimeter and volume in cubic centimeters, pressure times volume equals grams times cubic centimeters divided by square centimeters, which equals grams times centimeters—the equivalent of mass times distance moved, or work. Thus, a plot of pressure times volume for a single contraction of a ventricle yields a pressure-volume loop whose area is proportional to the mechanical work done by that ventricle.

Figure 12-13 illustrates pressure-volume loops for the right and left ventricles of a mammalian heart. The two ventricles eject equal volumes of blood, but the pressures generated in the pulmonary circuit (right ventricle) are much lower; consequently, the mechanical work done by the right ventricle is much less than that done by the left ventricle. As described in the previous section, blood is ejected from the ventricle when the intraventricular pressure exceeds the arterial pressure. If the arterial pressure is elevated, more mechanical work must be done by the heart to raise the intraventricular pressure enough to maintain stroke volume at the original level. This demand, of course, means that there is an extra strain on the heart if blood pressure is high.

Not all energy expended by the heart appears as changes in pressure and flow. Some energy is expended to overcome frictional forces within the myocardium, and more is dissipated as heat. The mechanical work done by the heart, expressed as a fraction of the total energy expended, is termed the efficiency of contraction. The mechanical work done can be determined from measurements of pressure and flow and converted into milliliters of O_2 consumed. This mea-

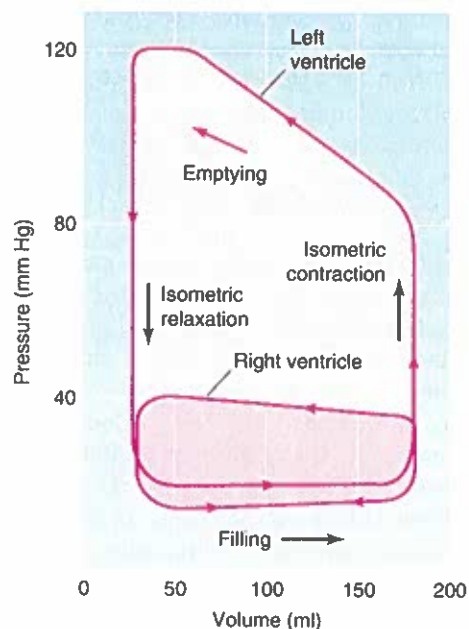


Figure 12-13 The area of a ventricular pressure-volume loop is proportional to the mechanical work done by a ventricle in one cardiac cycle. Shown here are loops for the right ventricle (red) and left ventricle (pink) of the mammalian heart. Once around a loop in a counterclockwise direction is equivalent to one heartbeat. Ventricular filling occurs at low pressure; pressure increases sharply only when the ventricles contract (the sharp upswing on the right-hand side of each loop). Ventricular volume decreases as blood flows into the arterial system, and ventricular pressure falls rapidly as the ventricle relaxes. Filling then begins again. Note that although the volume changes in both ventricles are similar, the pressure changes are much larger in the left ventricle than in the right one. Therefore, the left ventricle has a larger loop and hence does more mechanical work than the right ventricle.

sure, in turn, can be expressed as a fraction of the total O_2 uptake by the heart in order to measure the efficiency of contraction. In fact, no more than 10%–15% of the total energy expended by the heart appears as mechanical work.

Energy is expended to increase tension and raise blood pressure within the heart. According to Laplace's law, the relationship between wall tension and pressure in a hollow structure is related to the radius of curvature of the wall. If the structure is a sphere, then

$$P = \frac{2y}{R} \quad (12-1)$$

where P is the transmural pressure (the pressure difference across the wall of the sphere), y is the wall tension, and R is the radius of the sphere. According to this relation, a large heart must generate twice the wall tension of a heart half its size to develop a similar pressure. Thus, more energy must be expended by larger hearts in developing pressure, and we might expect a larger ratio of muscle mass to total heart volume in large hearts. Hearts are not, of course, perfect spheres, but have a complex gross and microscopic morphology; nevertheless, Laplace's law serves as a general guide. The energy expended in ejecting a given quantity of blood from the heart depends on the efficiency of contraction, the pressures developed, and the size and shape of the heart.

Coronary circulation

The coronary circulation supplies nutrients and oxygen to the heart. The blood supply to the heart is extensive—cardiac muscle has a much higher capillary density and more mitochondria than most skeletal muscles. It also has a high myoglobin content, which accounts for the typical red color of the heart. The blood flowing through the chambers of the heart supplies nutrients to the inner spongy layer of the heart in many fishes and amphibians, but even in these animals, an additional coronary supply is necessary to deliver oxygen and other substrates to the outer, denser regions of the heart wall. In general, hearts can use a wide variety of nutrients, including fatty acids, glucose, and lactate; the particular substrate used is determined largely by availability.

However, the heart relies primarily on aerobic pathways to generate energy, so it is very dependent on a continuous oxygen supply. An increase in cardiac activity depends on increased metabolism in the heart, which in turn requires increased coronary flow. Adenosine is a key metabolite in maintaining the relationship between coronary flow and cardiac activity. Adenosine and other local metabolic products cause dilation of coronary vessels and, therefore, increased coronary flow. The formation and release of adenosine increases with increased metabolism or during myocardial hypoxia (a drop in oxygen level). Increased adenosine levels also reduce heart rate and, therefore, energy expenditure. Thus the level of adenosine mediates the

match between energy expenditure and supply. Sympathetic stimulation is a second, but less important, mechanism of increasing coronary flow. Circulating catecholamines increase cardiac contractility and cause coronary vasodilation mediated via β_1 -adrenoreceptors.



Mammalian hearts have an extensive coronary circulation throughout the myocardium, but in some fish hearts, the coronary circulation is restricted to the epicardium, the external lining of the heart. What would be the consequences of this differing organization for the nutrition of the heart?

The pericardium

The heart is surrounded by a connective tissue membrane called the pericardium. The magnitude of the pressure changes within the pericardial cavity depend on the rigidity of the pericardium and on the magnitude and rate of change of the heart volume. In some animals, the pericardium is thin and flexible (compliant), in which case pressure changes within the pericardial cavity during each heartbeat are negligible. In others, such as sharks, the pericardium is quite rigid (noncompliant), in which case the intrapericardial pressure oscillates during each heartbeat.

The compliant pericardium enveloping the mammalian heart is formed of two layers, an outer fibrous layer and an inner serous layer. The serous layer is dou-

We'll see the pericardium in toad lab

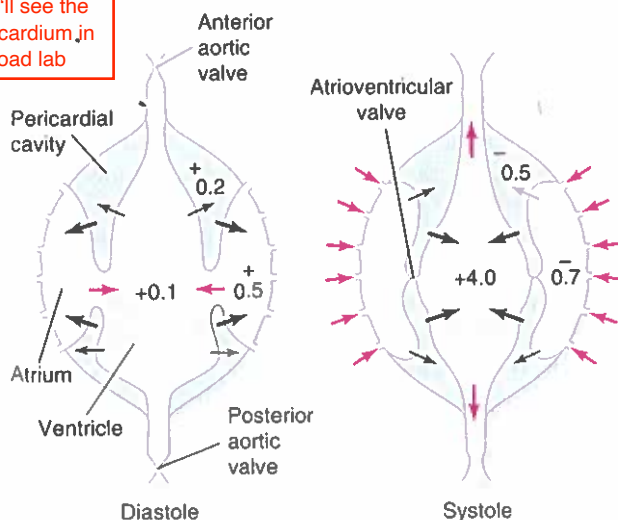
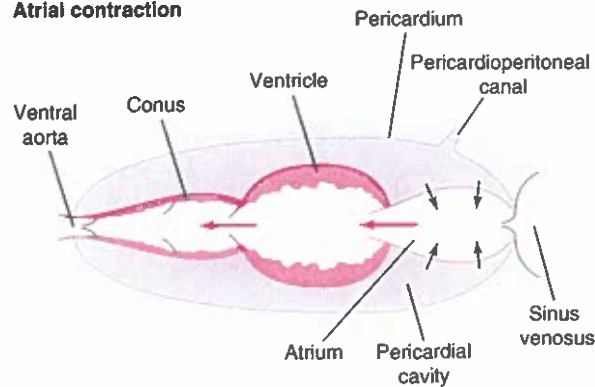
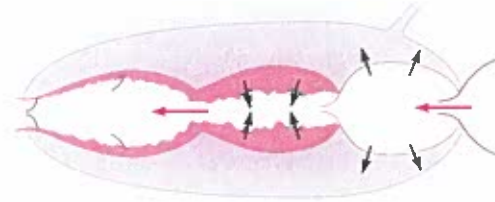


Figure 12-14 In the heart of the bivalve mollusk *Anodonta*, which has a noncompliant pericardium, ventricular contraction not only ejects blood, but also reduces pressure in the pericardial cavity, thus enhancing filling of the atria. Numbers are pressures in centimeters of seawater, which are expressed relative to ambient pressures. Large black arrows indicate movements of the walls of contracting chambers; small black arrows indicate movements of the walls of relaxing chambers. The red arrows indicate the direction of blood flow. [Adapted from Brand, 1972.]

Atrial contraction



Ventricular contraction



Conal contraction

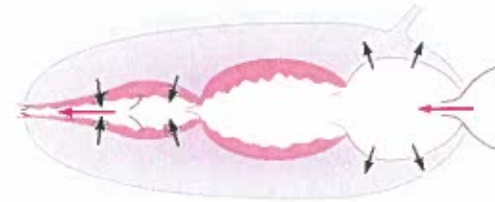


Figure 12-15 Because the elasmobranch heart is contained within a noncompliant pericardium, contraction of the ventricle reduces pressure in the pericardial cavity and assists atrial filling. In some elasmobranchs, fluid loss via the pericardioperitoneal canal during exercise, feeding, and coughing leads to an increase in heart size and stroke volume. Black arrows indicate the direction of wall movement during muscle contraction or relaxation; red arrows indicate the direction of blood flow.

ble, forming the inner lining of the pericardial cavity and the outer layer of the heart itself (the epicardium). In mammals, the serous layer secretes a fluid that acts as a lubricant, facilitating movement of the heart.

Crustaceans and bivalve mollusks have a noncompliant pericardium. In these animals, contraction of the ventricle reduces pressure in the pericardial cavity and enhances flow into the atria from the venous system (Figure 12-14). Thus, tension generated in the ventricular wall is utilized both to eject blood into the arterial system and to draw blood into the atria from the venous system.

The pericardia of elasmobranchs (sharks) and lungfishes are also noncompliant, whereas those of bony fishes are compliant. The elasmobranch heart consists of four chambers—sinus venosus, atrium, ventricle, and conus—all contained within a rigid pericardium (Figure 12-15). The reduction in intrapericardial

pressure that occurs during ventricular contraction in elasmobranchs produces a suction that helps expand the atrium and thereby increases venous return to the heart. If the pericardial cavity is opened, cardiac output is reduced; hence the increased venous return caused by reduced pericardial pressure must be important in augmenting cardiac output. In some elasmobranchs, a pericardioperitoneal canal exists between the pericardial and peritoneal cavities. There is little or no fluid flow through this canal in resting elasmobranchs, but during exercise, coughing, or feeding, the loss of fluid from the pericardial cavity via the canal causes an increase in heart size and stroke volume. This fluid is slowly replaced by plasma ultrafiltrate. A thin, flexible pericardium, although protective, has little effect on cardiac output, whereas the more rigid pericardium of sharks, and the possible variations in pericardial fluid volume arising from anatomic elaborations of the pericardium, can have a marked effect on cardiac output.

End

Vertebrate Hearts:

Comparative Functional Morphology

A comparative analysis of vertebrate circulatory systems produces insights into the relationships between heart structure and function. Numerous cardiovascular differences distinguish air-breathing vertebrates from those that do not breathe air. Among the air-breathing vertebrates, there is considerable variety in the extent to which the systemic and pulmonary circuits are separated.

The pulmonary circuit of birds and mammals is maintained at much lower pressures than is the systemic circuit. This difference in pressure is possible because birds and mammals have two series of heart chambers in parallel. The left side of the heart ejects blood into the systemic circulation, and the right side ejects blood into the pulmonary circulation (see Figure 12-3). The advantage of maintaining a high blood pressure is that it allows rapid transit times and rapid changes in blood flow through small-diameter capillaries. However, when the difference in pressure across a vessel wall (i.e., the transmural pressure) is high, fluid filters across the capillary wall; as a result, extensive lymphatic drainage of the tissues is necessary. In the mammalian lung, capillary flow can be maintained by relatively low input pressures, reducing the requirements for lymphatic drainage and avoiding the formation of large extracellular fluid spaces that could increase diffusion distances between blood and air and impair the gas transfer capacity of the lung. One advantage of a divided heart is that blood flow to the body and the lungs can be maintained with different input pressures. A disadvantage of a completely divided heart is that in order to avoid shifts in blood volume from the systemic to the pulmonary circuit, or vice versa, cardiac output must be the same in both sides of the heart, independent of the requirements in the two circuits.

In contrast, lungfishes, amphibians, reptiles, bird embryos, and fetal mammals have either an undivided

ventricle or some other mechanism that allows the shunting of blood from one circuit to the other. These shunts usually result in the movement of blood from the right (respiratory or pulmonary) to the left (systemic) side of the heart during periods of reduced gas transfer in the lung. At such times, blood returning from the body is shunted away from the lung to the left side of the heart and once again ejected into the systemic circuit, bypassing the lungs. In lungfishes, amphibians, and reptiles, flow to the lungs is commonly reduced during prolonged dives, during which gas transfer occurs across the skin, or oxygen stores in the body are used. Blood flow to the lungs is also reduced during development within the mother (mammals) or egg (birds), before the lungs become fully functional in gas exchange. Although a single undivided ventricle permits variations in the ratio of flows to the respiratory and systemic circuits, the same pressure must be maintained on both sides of the heart.

Water-breathing fishes

The hearts of water-breathing fishes, including elasmobranchs and most bony fishes (teleosts), consist of four chambers in series. All the chambers are contractile except the elastic bulbus of teleosts. Unidirectional flow of blood through the heart is maintained by valves at the sinoatrial and atrioventricular junctions and at the exit of the ventricle.

In elasmobranchs, the exit from the ventricle to the conus is guarded by a pair of flap valves, and there are from two to seven pairs of valves along the length of the conus, depending on the species (see Figure 12-15). Conus length is variable among species; in general, more valves are found in those species with a longer conus. Just before ventricular contraction, all valves except the set most distal to the ventricle are open; that is, the conus and the ventricle are interconnected, but a closed valve at the exit of the conus maintains a pressure difference between the conus and the ventral aorta. During atrial contraction, both the ventricle and the conus are filled with blood. Ventricular contraction in elasmobranchs does not have an isovolumic phase, as in mammals, because at the onset of contraction blood is moved from the ventricle into the conus. Pressure rises in the ventricle and conus and eventually exceeds that in the ventral aorta. The distal valves then open, and blood is ejected into the aorta. During conal contraction, which begins after the onset of ventricular contraction, the proximal valves close, preventing reflux of blood into the ventricle as it relaxes. Conal contraction proceeds relatively slowly away from the heart toward the aorta; each set of valves closes in turn to prevent backflow of blood.

As illustrated in Figure 12-16a, blood pumped by the heart in typical water-breathing fishes passes first through the gill (respiratory) circulation and then into a dorsal aorta that supplies the rest of the body (systemic circulation). Thus, the respiratory and systemic circuits